Fostering STEM Engagement in Spatial Reasoning from Parent-Child Interaction: A Remote Study

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Parent-directedness during an at-home spatial reasoning activity reduces 3-to-5-year-old children's engagement in related challenges.

Background

- Parents support children's **STEM learning** through everyday activities before they begin formal education[1-3]
- Parents also foster children's interest through the *autonomy* they provide during STEM activities, which promotes children's **STEM engagement**[4]
- We investigate how parent-child interaction, focused on autonomy support during an **at-home** activity focused on spatial relations, relates to preschoolers' engagement in spatial reasoning challenges
- We predict that when parents provide children with ***less autonomy*** during the activity**, children's engagement would be reduced** during related STEM challenges.

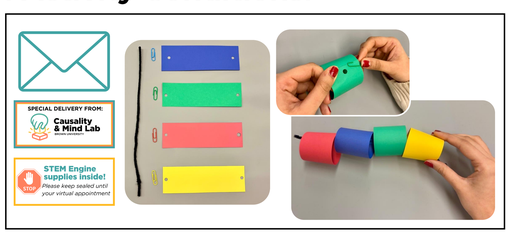
Methods

Participants: 3- to 5-year-olds (N=160, 85 girls, M_{age}= 53.87 months, SD =10.01)

PREREGISTERED

Parent-Child STEM Activity Conditions:

- **Trains** (n = 100): "Make a train" kit mailed to families. Constructed a train using craft materials. Followed by reflections.



- **Transfer** (n = 30): Families completed an alternative STEM activity that demonstrates how soap breaks surface tension in water.
- **Baseline** (n = 30) No parent-child activity

Coding PCI Style[4-5] for *Trains* Condition

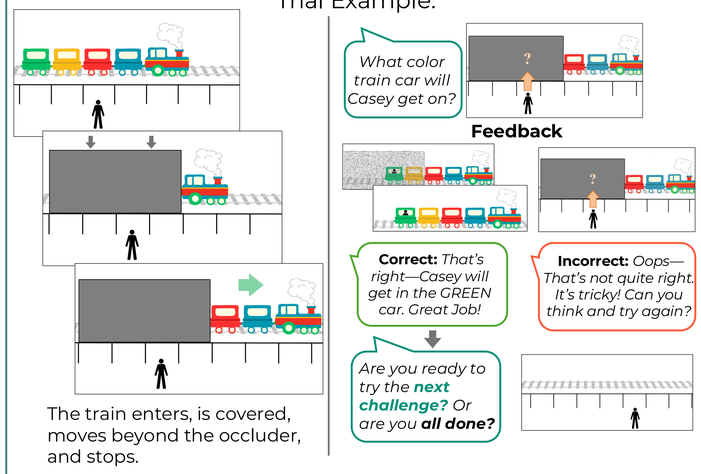
Parent-Directed	Jointly Directed	Child-Directed
Parents set goals and often take over tasks for their child	Parents supported, collaborative goal-setting	Parents let children set goals for themselves

Additional measures:

Reflections: **(1)** Causal Language and **(2)** MLU	**IC** Inhibitory control (Day/Night Stroop)[6]	**WM** Working Memory - (Digit Span)[7]	**ToM** Theory of Mind Scales[8]	**Surgency and Control** Temperament subscales (Parent report)[9]	Parent Attitudes Towards Science[10]

Spatial Reasoning Challenges

Trial Example:



Trials progressively increased in difficulty.

We measured:

- **Engagement Score:** number of challenges children participated in voluntarily (out of a possible 10)
- **Performance Score:** proportion of challenges solved correctly.

Results

- On average, children chose to complete 5.62 challenges (SD = 3.58, Range: 1-10).
- Children's engagement and performance scores were negatively correlated, $r_s(158) = -.57$, $p < .001$.

Correlations among engagement and performance scores, age, and control measures:

	Age	WM	IC	ToM	Parent Questionnaire: Surgency	Parent Questionnaire: Control	Parent Questionnaire: Parent Science Attitudes
Engagement	**.19***	**.25****	.11	**.20***	-.14	**.20***	.02
Performance	**.22****	**.19***	**.22****	**.18***	.05	-.09	.02

*<.05, **<.01

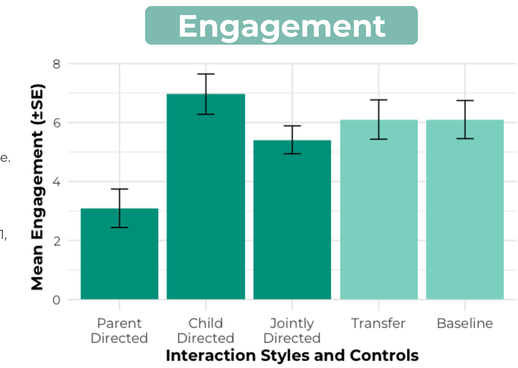
Results: Model Analyses

Children in the parent-directed group completed *fewer* challenges than those in all other interaction style groups and control conditions.

GLM, Engagement ~ Age (months) + Condition/PCI + Performance. Plot Displays SE Error bars.

- Condition effect is present in light of a main effect of **age**, B = 0.06, Wald χ^2 (1) = 11.38, $p < .001$, and **performance**, B = -6.67, Wald χ^2 (1) = 49.06, $p < .001$.

Condition effect remains significant in an exploratory model including WM, TOM, and Temperament.

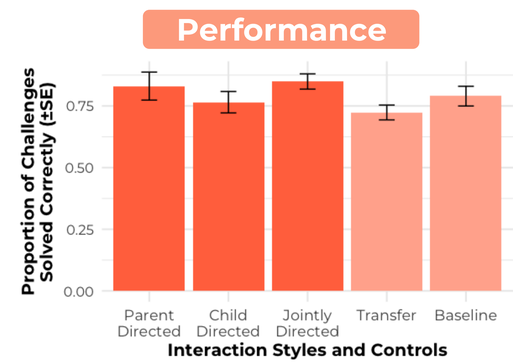


There were no significant condition differences in children's performance scores.

GLM, Performance ~ Age (months) + Condition/PCI + Engagement. All comparisons between parent-directed and other groups, n.s.

- Children were more accurate with **age**, B = 0.07, Wald χ^2 (1) = 16.02, p < .001.
- More accurate children also tended to have *lower* **engagement**, B = -0.45, Wald χ^2 (1) = 65.25, p < .001. Plot Displays SE Error bars.

An exploratory model including WM, IC, and ToM as covariates was not significant.



PCI Style Comparisons

- Children in the **child-directed group were significantly older** than those in the parent-directed and jointly-directed groups.
- We reproduced our analyses only on the PCI groups, including causal language, MLU, and our cognitive measures.
  - Only the dyad's **PCI style** significantly predicted their **engagement** score, Wald $\chi^2(2) = 7.75$, p = .02, Parent-directed < Child and Jointly-directed.
  - Overall model on **performance** was n.s., but suggested a main effect of **WM**.

Discussion

- When parents are **overly directive** in setting goals for their preschool-age child during at-home STEM activities, children are **less engaged** in related activities, following this interaction.
- This work contributes to a greater understanding of motivational influences on children's engagement with spatial reasoning, an important precursor to engineering, before children are introduced to engineering concepts in the classroom.

References

[1] Braham et al. (2018). *Dev. Psych.* **[2]** Tamis-LeMonda et al. (2004). *Child Dev.* **[3]** Morris et al. (2023). *J. Applied Dev. Psych* **[4]** Sobel (2023). *Cur Dir Psych Sci* **[5]** Fung & Callanan (2013). Poster at SRCD **[6]** Gerstadt et al. (1994). *Cognition* **[7]** Wechsler (2002). WPPSI-III. **[8]** Wellman & Liu (2004). *Child Development* **[9]** Putnam & Rothbart (2009). *J. Personality Assessment.* **[10]** Richard (2019). Dissertation.

Acknowledgements and Contact

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